

FERMION FIELD THEORY @ TREE LEVEL

→ Continuous symmetries + conserved currents

- Punchline: If a set of infinitesimal transformations leaves $\mathcal{L}$ invariant, i.e., $\mathcal{L}$ has a continuous symmetry $\Rightarrow$ a conserved charge.

↳ Derivation:

- Scalar $\phi(x)$, $\mathcal{L}(x) = \mathcal{L}(\phi(x), \partial_\mu \phi(x))$

Make an infinitesimal transformation $\phi(x) \rightarrow \phi(x) + \delta\phi(x) \Rightarrow \mathcal{L}(x) \rightarrow \mathcal{L}(x) + \delta\mathcal{L}(x)$

- By chain rule, $\delta\mathcal{L}(x) = \frac{\partial\mathcal{L}(x)}{\partial\phi(x)} \delta\phi(x) + \frac{\partial\mathcal{L}(x)}{\partial(\partial_\mu\phi(x))} \partial_\mu \delta\phi(x)$

- By principle of least action, $\frac{\delta S}{\delta\phi(x)} = \frac{\partial\mathcal{L}(x)}{\partial\phi(x)} - \partial_\mu \left(\frac{\partial\mathcal{L}(x)}{\partial(\partial_\mu\phi(x))} \right) = 0$

$$\Rightarrow \frac{\partial\mathcal{L}(x)}{\partial\phi(x)} = \partial_\mu \left(\frac{\partial\mathcal{L}(x)}{\partial(\partial_\mu\phi(x))} \right) + \frac{\delta S}{\delta\phi(x)} \rightarrow \text{plug into first term of } \delta\mathcal{L} \text{ expansion}$$

$$\Rightarrow \delta\mathcal{L}(x) = \partial_\mu \left(\underbrace{\frac{\partial\mathcal{L}(x)}{\partial(\partial_\mu\phi(x))} \delta\phi(x)}_{\text{Noether current } j^\mu} \right) + \frac{\delta S}{\delta\phi(x)} \delta\phi(x) \Rightarrow \partial_\mu j^\mu = \delta\mathcal{L}(x) - \frac{\delta S}{\delta\phi(x)} \delta\phi(x)$$

- If $\phi(x)$ satisfies classical field eqns (E-L) $\Rightarrow \frac{\delta S}{\delta\phi} = 0 \Rightarrow \partial_\mu j^\mu = \delta\mathcal{L}(x)$

- If there is a set of infinitesimal transformations that leaves the Lagrangian invariant $\Rightarrow \delta\mathcal{L} = 0 \rightarrow \mathcal{L}$ has a continuous symmetry

$$\Rightarrow \partial_\mu j^\mu = 0 \rightarrow \text{conserved current} \rightarrow \underbrace{\frac{\partial j^0(x)}{\partial t}}_{\text{charge density}} + \underbrace{\nabla \cdot \vec{j}(x)}_{\text{current density}} = 0 \rightarrow \text{continuity eqn.}$$

- Noether charge: $Q \equiv \int d^3x j^0(x)$

↳ Example: complex scalar

$$\mathcal{L} = -\partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi - \frac{1}{4} \lambda (\phi^\dagger \phi)^2$$

- Continuous global $U(1)$ symmetry: $\phi \rightarrow e^{-i\alpha} \phi$ (equivalently, can choose $SO(2)$ transformation)
infinitesimal: $\phi(x) \rightarrow \phi(x) - i\alpha \phi(x)$, $\phi^\dagger(x) \rightarrow \phi^\dagger(x) + i\alpha \phi^\dagger(x)$

- Noether current: $j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)} \delta\phi^\dagger = (-\partial_\mu \phi^\dagger)(-i\alpha\phi) + (-\partial_\mu \phi)(i\alpha\phi^\dagger)$
 $= \alpha \text{Im}(\phi^\dagger \overleftrightarrow{\partial}^\mu \phi) \Rightarrow j^\mu = \text{Im}(\phi^\dagger \overleftrightarrow{\partial}^\mu \phi)$

- Noether charge $Q = \int d^3x \text{Im}(\phi^\dagger \overleftrightarrow{\partial}^0 \phi) = \int d^3x [a^\dagger(\vec{k}) a(\vec{k}) - b(\vec{k}) b^\dagger(\vec{k})]$ (after mode exp.)

↳ Noether's theorem in QFT

- By imposing $\delta\mathcal{L}(x) = 0$ when $\phi(x) \rightarrow \phi(x) + \delta\phi(x)$,

and taking n functional derivatives wrt J , we get

$$0 = i \langle 0 | T \frac{\delta S}{\delta\phi_a(x)} \phi_a(x_1) \dots \phi_a(x_n) | 0 \rangle + \sum_{j=1}^n \langle 0 | T \phi_a(x_1) \dots \delta\phi_a(x_j) \delta^4(x-x_j) \dots \phi_a(x_n) | 0 \rangle$$

↳ Schwinger-Dyson eqns for a set $\{\phi_a(x)\}$

- Imposing a symmetry $\delta\mathcal{L}(x) = 0$, and using $\partial_\mu j^\mu(x) = \delta\mathcal{L} - \frac{\delta S}{\delta\phi_a(x)} \delta\phi_a(x)$

$$0 = \partial_\mu \langle 0 | T j^\mu(x) \phi_a(x_1) \dots \phi_a(x_n) | 0 \rangle + i \sum_{j=1}^n \langle 0 | T \phi_a(x_1) \dots \delta\phi_a(x_j) \delta^4(x-x_j) \dots \phi_a(x_n) | 0 \rangle$$

↳ Ward identity $\rightarrow$ Noether current is conserved in QFT, upto contact terms.

Two component Weyl Spinors:

- Punchline: By considering the way fields transform under Lorentz transformations, we come across spinors, an irreducible representation of the Lorentz group.

→ Scalars → defined as being invariant under LTs. How do other objects transform under LTs?

→ Lorentz group Lie algebra:

$$[J_i, J_j] = i\epsilon_{ijk} J_k, [J_i, K_j] = i\epsilon_{ijk} K_k, [K_i, K_j] = -i\epsilon_{ijk} J_k$$

3 $(2j+1) \times (2j+1)$ Hermitian matrices obey this relation.

For $j = 1/2 \Rightarrow J_i = \sigma_i$

$j = 0, 1/2, 1, \dots \Rightarrow$ we can find all reps of $SU(2)$ algebra → each rep consists of these 3 matrices.

- Considering all 3 commutators → graduate from $SU(2)$ to $SL(2, \mathbb{C})$

$$\text{Define } \begin{cases} N_i = \frac{1}{2} (J_i - iK_i) \\ N_i^\dagger = \frac{1}{2} (J_i + iK_i) \end{cases} \Rightarrow \begin{cases} [N_i, N_j] = i\epsilon_{ijk} N_k \\ [N_i^\dagger, N_j^\dagger] = i\epsilon_{ijk} N_k^\dagger \\ [N_i, N_j^\dagger] = 0 \end{cases}$$

each give 2 independent $SU(2)$ Lie algebras.

$so(1,3) \cong su(2) \oplus su(2)$
Lorentz g.p. ↑ direct product.

- we identify these reps by

$$(j, j') \text{ or } (2j+1, 2j'+1), j, j' = 0, 1/2, 1, \dots$$

e.g. scalar: $(0,0)$

spinors: $(1/2, 0)$ or $(0, 1/2)$

→ Weyl spinors:

- An arbitrary field ϕ transforms under LTs as: $\phi'(x^\mu) = L(\Lambda) \phi(\Lambda^{-1} x^\mu)$

$$L = \exp\left(\frac{i}{2} \omega_{\mu\nu} M^{\mu\nu}\right) \approx \mathbb{1} - i \vec{\theta} \cdot \vec{J} - i \vec{\xi} \cdot \vec{K}$$

- In the $(1/2, 0)$ or $(0, 1/2)$ reps, the fields are called spinors

Left Handed: $(1/2, 0)$ rep, ψ_a

Right Handed: $(0, 1/2)$ rep, $\psi_a^\dagger = (\psi_a(x))^\dagger$

distinguish between dotted and undotted indices → cannot contract $a, \dot{a}$ to make a Lorentz scalar. $\psi^\dagger \psi$

→ Spinor Lorentz Transform:

$$(1/2, 0) \otimes (0, 1/2) \neq (0, 0)$$

$$\psi_a \rightarrow L_a^b \psi_b \Rightarrow \psi_a^\dagger \rightarrow (L^*)_{\dot{a}}^{\dot{b}} \psi_{\dot{b}}^\dagger$$

→ Raising and lowering spinor indices:

- Spinor metric tensor ϵ (anti-symmetric) → $\epsilon^{12} = \epsilon_{21} = +1, \epsilon^{21} = \epsilon_{12} = -1$

$$\psi_a = \epsilon_{ab} \psi^b, \psi^a = \epsilon^{ab} \psi_b \rightarrow \epsilon^{\dot{a}\dot{b}} = (\epsilon^{ab})^*, \epsilon_{\dot{a}\dot{b}} = (\epsilon_{ab})^*$$

$$\psi_{\dot{a}}^\dagger = \epsilon_{\dot{a}\dot{b}} \psi_{\dot{b}}^\dagger, \psi^{\dagger\dot{a}} = \epsilon^{\dot{a}\dot{b}} \psi_{\dot{b}}^\dagger$$

→ $(\sigma^\mu)_{a\dot{b}}$ and $(\bar{\sigma}^\mu)^{\dot{a}b}$

$$\sigma^\mu = (\mathbb{1}, \vec{\sigma}), \bar{\sigma}^\mu = (\mathbb{1}, -\vec{\sigma}) \rightarrow \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Vector and spinor indices raised and lowered by $g_{\mu\nu}, \epsilon_{ab}$

$$\sigma^\mu_{a\dot{a}} = \epsilon_{ab} \epsilon_{\dot{a}\dot{b}} (\bar{\sigma}^\mu)^{\dot{b}b}$$

$$(\bar{\sigma}^\mu)^{\dot{a}a} = \epsilon^{ab} \epsilon^{\dot{a}\dot{b}} \sigma^\mu_{\dot{b}b}$$

→ Constructing Lorentz tensors from fermion spinors

- Fermion bilinears: $\chi\eta = \chi^a \eta_a, \chi^\dagger \eta^\dagger = \chi_a^\dagger \eta^{\dagger a}$ (rank-0)

$$\chi^\dagger \bar{\sigma}^\mu \eta = \chi_a^\dagger (\bar{\sigma}^\mu)^{\dot{a}b} \eta_b$$

$$\chi \sigma^\mu \eta^\dagger = \chi^a (\sigma^\mu)_{a\dot{b}} \eta^{\dagger\dot{b}}$$

→ Fermion anticommutation:

$$\chi_a(x) \psi_b(y) = -\psi_b(y) \chi_a(x)$$

→ Spinor Lagrangian

- Free fermion: $\mathcal{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \frac{1}{2} m \psi \psi - \frac{1}{2} m^* \psi^\dagger \psi^\dagger \rightarrow$ Lorentz invariant
Hermitian (upto total derivative)
Quadratic in $\psi, \psi^\dagger$
- com: By E-L eqns: $-i\bar{\sigma}^\mu \partial_\mu \psi^\dagger + m\psi = 0$
 $-i\bar{\sigma}^\mu \partial_\mu \psi + m\psi^\dagger = 0$

→ Dirac equation:

- Combine both coms into a matrix eqn: $\begin{pmatrix} m & -i\bar{\sigma}^\mu \partial_\mu \\ -i\bar{\sigma}^\mu \partial_\mu & m \end{pmatrix} \begin{pmatrix} \psi \\ \psi^\dagger \end{pmatrix} = 0 \rightarrow$ new spinor rep
- Define: $\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \rightarrow \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu} \rightarrow (-i\gamma^\mu \partial_\mu + m) \Psi = 0 \rightarrow$ Dirac eqn.

→ Dirac spinors:

- Consider 2 left handed weyl spinors ψ_1, ψ_2 w/ same mass
 $\mathcal{L} = i\psi_i^\dagger \bar{\sigma}^\mu \partial_\mu \psi_i - \frac{1}{2} m \psi_i \psi_i - \frac{1}{2} m \psi_i^\dagger \psi_i^\dagger, i=1,2 \rightarrow$ Rewrite: $\begin{cases} \chi = \frac{1}{\sqrt{2}} (\psi_1 + i\psi_2) \\ \eta = \frac{1}{\sqrt{2}} (\psi_1 - i\psi_2) \end{cases}$
- $\Rightarrow \mathcal{L} = i\chi^\dagger \bar{\sigma}^\mu \partial_\mu \chi + i\eta^\dagger \bar{\sigma}^\mu \partial_\mu \eta - m\chi\eta - m\eta^\dagger\chi^\dagger \rightarrow$ invariant under global $U(1): \chi \rightarrow e^{-i\alpha}\chi, \eta \rightarrow e^{i\alpha}\eta$
- Dirac spinor $\Psi = \begin{pmatrix} \chi \\ \eta^\dagger \end{pmatrix} \rightarrow \bar{\Psi} = \Psi^\dagger \gamma^0 = (\eta, \chi^\dagger) \rightarrow \mathcal{L} = i\bar{\Psi} \gamma^\mu \partial_\mu \Psi - m\bar{\Psi} \Psi$
 $U(1): \Psi \rightarrow e^{i\alpha}\Psi, \bar{\Psi} \rightarrow e^{-i\alpha}\bar{\Psi}$

- Noether current: $j^\mu = \bar{\Psi} \gamma^\mu \Psi = \chi^\dagger \bar{\sigma}^\mu \chi - \eta^\dagger \bar{\sigma}^\mu \eta$

→ Projection operator:

- Takes us from Dirac to weyl rep: $\gamma_5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
- $P_L = \frac{1}{2}(1 - \gamma_5) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, P_R = \frac{1}{2}(1 + \gamma_5) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow$ project out left- and right-handed components.
- $P_L \Psi = \begin{pmatrix} \chi \\ 0 \end{pmatrix}, P_R \Psi = \begin{pmatrix} 0 \\ \eta \end{pmatrix} \rightarrow \gamma_5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \frac{-i}{24} \epsilon_{\mu\nu\rho\sigma} \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma$

→ Canonical Quantization of spinors:

- Left Handed Weyl field $\psi \rightarrow \mathcal{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \frac{1}{2} m (\psi\psi + \psi^\dagger\psi^\dagger)$
- Canonical conjugate momentum: $\Pi^a(x) \equiv \frac{\partial \mathcal{L}}{\partial \dot{\psi}_a(x)} = i\psi_a^\dagger(x) \bar{\sigma}^{0a}$

- Hamiltonian: $\mathcal{H} = \Pi^a \partial_0 \psi_a - \mathcal{L} = -i\psi^\dagger \bar{\sigma}^i \partial_i \psi + \frac{1}{2} m (\psi\psi + \psi^\dagger\psi^\dagger)$

- Anti-commutation relations: $\{\psi_a(\vec{x}, t), \psi_b(\vec{y}, t)\} = 0$

- Four-component notation: $\{\psi_a(\vec{x}, t), \Pi^c(\vec{y}, t)\} = i\delta_a^c \delta^3(\vec{x} - \vec{y})$
 $\{\psi_a(\vec{x}, t), \psi_b^\dagger(\vec{y}, t)\} \bar{\sigma}^{0a} = \delta_a^b \delta^3(\vec{x} - \vec{y})$

- $\{\psi_a(\vec{x}, t), \bar{\psi}_b(\vec{y}, t)\} = (\gamma^0)_{ab} \delta^3(\vec{x} - \vec{y}) \Rightarrow \{\psi_a(\vec{x}, t), \psi_b^\dagger(\vec{y}, t)\} = \sigma_{ab}^0 \delta^3(\vec{x} - \vec{y})$

→ Mode expansion:

$$\psi \equiv a_\mu \gamma^\mu$$

- Dirac eqn: $(-i\gamma^\mu \partial_\mu + m)\Psi = (-i\not{\partial} + m)\Psi = 0 \Rightarrow 0 = (\not{\partial} + m)(-i\not{\partial} + m)\Psi = (\not{\partial}\not{\partial} + m^2)\Psi$

$\Rightarrow \Psi$ can be written as a superposition of waves. $= (-\not{\partial} + m^2)\Psi \rightarrow k^0$

- $\Psi(x) = \sum_{s=\pm} \int d^3p [b_s(\vec{p}) u_s(\vec{p}) e^{i\vec{p}\cdot\vec{x}} + d_s^\dagger(\vec{p}) v_s(\vec{p}) e^{-i\vec{p}\cdot\vec{x}}]$

- $(-\not{\partial} + m)\Psi(x) = (\not{p} + m)u_s(\vec{p}) e^{i\vec{p}\cdot\vec{x}} + (-\not{p} + m)v_s(\vec{p}) e^{-i\vec{p}\cdot\vec{x}} = 0$

$$\Rightarrow (\not{p} + m)u_s(\vec{p}) = 0$$

$$(-\not{p} + m)v_s(\vec{p}) = 0$$

→ Spinor Technology:

↳ Explicit forms for u, v

• $\begin{cases} (\not{p} + m) u_s(\vec{p}) = 0 \\ (-\not{p} + m) v_s(\vec{p}) = 0 \end{cases} \rightarrow \text{rest frame: } m \neq 0, \vec{p} = 0 \rightarrow \not{p} = -m \gamma^0 = -m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

• u, v still not uniquely defined $\rightarrow$ choose as eigenstates of $S_z = \frac{i}{2} [\gamma^1, \gamma^2]$

$S_z u_{\pm}(\vec{0}) = \pm \frac{1}{2} u_{\pm}(\vec{0}) \Rightarrow u_+(\vec{0}) = \sqrt{m} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, u_-(\vec{0}) = \sqrt{m} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \sigma_3 & 0 \\ 0 & \sigma_3 \end{pmatrix}$

$S_z v_{\pm}(\vec{0}) = \mp \frac{1}{2} v_{\pm}(\vec{0})$
 $v_+(\vec{0}) = \sqrt{m} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, v_-(\vec{0}) = \sqrt{m} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

↳ Spinor identities

• Bilinears: $\bar{u}_s'(\vec{p}) u_s(\vec{p}) = \bar{u}_s'(\vec{0}) \underbrace{\exp(-i\eta \vec{p} \cdot \vec{\alpha}) \exp(i\eta \vec{p} \cdot \vec{\alpha})}_{\text{boost ops}} u_s(\vec{0}) = \bar{u}_s'(\vec{0}) u_s(\vec{0}) = 2m \delta_{ss'}$

$\bar{v}_s'(\vec{p}) v_s(\vec{p}) = -2m \delta_{ss'}$

$\bar{u}_s'(\vec{p}) v_s(\vec{p}) = 0, \bar{v}_s'(\vec{p}) u_s(\vec{p}) = 0$

• Gordon identities:

$2m \bar{u}_s'(\vec{p}') \gamma^{\mu} u_s(\vec{p}) = \bar{u}_s'(\vec{p}') [(p' + p)^{\mu} - 2i S^{\mu\nu} (p' - p)_{\nu}] u_s(\vec{p}), S^{\mu\nu} = \frac{i}{4} [\gamma^{\mu}, \gamma^{\nu}]$

$-2m \bar{v}_s'(\vec{p}') \gamma^{\mu} v_s(\vec{p}) = \bar{v}_s'(\vec{p}') [(p' + p)^{\mu} - 2i S^{\mu\nu} (p' - p)_{\nu}] v_s(\vec{p})$

When $p' = p$

$\bar{u}_s'(\vec{p}) \gamma^{\mu} u_s(\vec{p}) = 2p^{\mu} \delta_{ss'}, \bar{v}_s'(\vec{p}) \gamma^{\mu} v_s(\vec{p}) = 2p^{\mu} \delta_{ss'}$

• $\sum_s u_s(\vec{p}) \bar{u}_s(\vec{p}) = \not{p} + m$

$\sum_s v_s(\vec{p}) \bar{v}_s(\vec{p}) = -\not{p} - m$

→ Free Fermion propagator:

• using the mode expansion of Ψ , we can express $b_s(\vec{p}), d_s^{\dagger}(\vec{p})$ in terms of the field, giving us anticommutators for the creation/annihilation ops.

$\{b_s(\vec{p}), b_{s'}^{\dagger}(\vec{p}')\} = (2\pi)^3 \delta^3(\vec{p} - \vec{p}') 2\omega \delta_{ss'}$
 $\{d_s(\vec{p}), d_{s'}^{\dagger}(\vec{p}')\} = (2\pi)^3 \delta^3(\vec{p} - \vec{p}') 2\omega \delta_{ss'}$ } all other relations vanish.

$H = \sum_{s=\pm} \int d\vec{p} \omega [b_s^{\dagger}(\vec{p}) b_s(\vec{p}) + d_s^{\dagger}(\vec{p}) d_s(\vec{p})]$ upto a const.

$Q = \sum_{s=\pm} \int d\vec{p} [b_s^{\dagger}(\vec{p}) b_s(\vec{p}) - d_s^{\dagger}(\vec{p}) d_s(\vec{p})]$

↳ Fermion propagator:

$S(x-y)_{\alpha\beta} \equiv i \langle 0 | T \psi_{\alpha}(x) \bar{\psi}_{\beta}(y) | 0 \rangle \rightarrow$ plug in mode expansion and evaluate

$S(x-y)_{\alpha\beta} = \int \frac{d^4 p}{(2\pi)^4} e^{ip(x-y)} \frac{(-\not{p} + m)_{\alpha\beta}}{p^2 + m^2 - i\epsilon} = i\theta(x^0 - y^0) \dots + i\theta(y^0 - x^0) \dots$
 ↑ contour integration

$\langle 0 | T \psi \psi | 0 \rangle = \langle 0 | T \bar{\psi} \bar{\psi} | 0 \rangle = 0$

→ Path integral for fermions:

↳ Free path integral: $Z_0(\bar{\eta}, \eta) = \int D\psi D\bar{\psi} e^{i \int d^4 x (\bar{\psi} \not{\partial} \psi + \bar{\psi} \eta + \bar{\eta} \psi)} = \exp[i \int d^4 x d^4 y \bar{\eta}(x) S(x-y) \eta(y)]$

↳ Correlation functions: $\langle 0 | T \psi_{\alpha_1}(x_1) \dots \bar{\psi}_{\beta_n}(y_n) | 0 \rangle = \left(\frac{1}{i} \frac{\delta}{\delta \bar{\eta}_{\alpha_1}(x_1)} \right) \dots \left(i \frac{\delta}{\delta \eta_{\beta_n}(y_n)} \right) \dots Z_0(\bar{\eta}, \eta) \Big|_{\bar{\eta}, \eta = 0}$
 (functional deriv. w/ anticommuting variable is also anticommuting)

↳ Interacting path integral: $Z_{\text{int}}(\bar{\psi}, \psi)$

$Z(\bar{\eta}, \eta) \propto \int D\psi D\bar{\psi} e^{i \int d^4 y \mathcal{L}_{\text{int}}(y)} e^{i \int d^4 y (\bar{\psi} \not{\partial} \psi + \bar{\psi} \eta + \bar{\eta} \psi)} = e^{i \int d^4 x \mathcal{L}_{\text{int}} \left(i \frac{\delta}{\delta \eta}, i \frac{\delta}{\delta \bar{\eta}} \right)} Z_0(\bar{\eta}, \eta)$
 $\mathcal{L}_{\text{int}} \left(i \frac{\delta}{\delta \eta}, i \frac{\delta}{\delta \bar{\eta}} \right)$

→ Fermionic LSZ formula

↳ Derived similarly to scalar:

$$\begin{aligned} \circ \langle f|i \rangle &= i^4 \int d^4x_1 d^4x_2 d^4x_3 d^4x_4 \times e^{-i p_1' x_1'} [\bar{u}_1, (-i \not{\partial}_1' + m)]_{\alpha_1'} \times e^{-i p_2' x_2'} [\bar{u}_2, (-i \not{\partial}_2' + m)]_{\alpha_2'} \\ &\times \langle 0 | T \psi_{\alpha_2'}(x_2') \psi_{\alpha_1'}(x_1') \bar{\psi}_{\alpha_3'}(x_3') \bar{\psi}_{\alpha_4'}(x_4') | 0 \rangle \\ &\times [(+i \not{\partial}_1 + m) u_1]_{\alpha_1} e^{i p_1 x_1} \times [(+i \not{\partial}_2 + m) u_2]_{\alpha_2} e^{i p_2 x_2} \end{aligned}$$

↳ Creation/annihilation ops should work the same in an interacting theory

$$\Rightarrow \langle 0 | \psi(x) | 0 \rangle = 0 \rightarrow \text{vev } 0$$

$$\begin{aligned} \langle p, s, + | \psi(x) | 0 \rangle &= 0 \rightarrow \langle p, s, - | \bar{\psi}(x) | 0 \rangle = 0 \\ \langle p, s, - | \psi(x) | 0 \rangle &= v_s(\vec{p}) e^{i p x} \\ \langle p, s, + | \bar{\psi}(x) | 0 \rangle &= \bar{u}_s(\vec{p}) e^{-i p x} \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} d^+ \text{ creates antiparticle} \\ b^+ \text{ creates particle} \end{array}$$

→ Dirac field Feynman Rules:

↳ Scalar-fermion interaction: $\mathcal{L}_{int} = y \bar{\psi} \psi \phi \rightarrow$ Yukawa interaction

$$\begin{aligned} \circ \text{Path integral: } Z(\eta, \bar{\eta}, J) &\propto \exp \left[i y \int d^4x \left(\frac{1}{i} \frac{\delta}{\delta \bar{\psi}(x)} \right) \left(\frac{i \delta}{\delta \eta(x)} \right) \left(\frac{-i \delta}{\delta J(x)} \right) \right] Z_0(\eta, \bar{\eta}, J) \\ S(x-y) &= \frac{\int d^4p}{(2\pi)^4} \frac{(-\not{p} + m) e^{i p (x-y)}}{p^2 + m^2 - i\epsilon} \rightarrow \text{fermion (mass } m) \\ \Delta(x-y) &= \frac{\int d^4k}{(2\pi)^4} \frac{e^{i k (x-y)}}{k^2 + M^2 - i\epsilon} \rightarrow \text{scalar (mass } M) \end{aligned}$$

$Z(\eta, \bar{\eta}, J) = \exp[iW(\eta, \bar{\eta}, J)] \times \exp \left[\int d^4x d^4y J(x) \Delta(x-y) J(y) \right]$
contributions from all connected non-vacuum diagrams

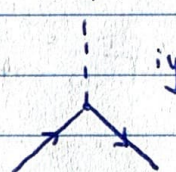
↳ Feynman rules:

- Find correlators by taking func. derivs.: $\langle 0 | T \psi(x) \bar{\psi}(y) \psi(z_1) \bar{\psi}(z_2) | 0 \rangle = \frac{1}{i} \frac{\delta}{\delta \bar{\eta}(x)} \frac{i \delta}{\delta \eta(y)} \frac{1}{i} \frac{\delta}{\delta \bar{\eta}(z_1)} \frac{i \delta}{\delta \eta(z_2)} Z(\eta, \bar{\eta}, J) \Big|_{\eta, \bar{\eta}, J=0}$
- Plug into LSZ formula to find $\langle f|i \rangle = (2\pi)^4 \delta^4(k_{in} - k_{out}) i T$
- Feynman rules $\rightarrow$ we work in p space.

external lines: (fermion)

- $\vec{p} \rightarrow$ incoming particle $u_s(\vec{p})$
- $\vec{p} \leftarrow$ incoming anti-particle $\bar{v}_s(\vec{p})$
- $\vec{p} \rightarrow$ outgoing particle $\bar{u}_s(\vec{p})$
- $\vec{p} \leftarrow$ outgoing anti-particle $v_s(\vec{p})$

vertex:



external lines: (scalar)

- $\vec{k} \rightarrow$ incoming scalar 1
- $\leftarrow \vec{k}$ outgoing scalar 1

propagators:

scalar $o \cdots o \rightarrow \frac{-i}{k^2 + m^2}$

fermion $o \cdots o \rightarrow \frac{-i(-\not{p} + m)}{p^2 + m^2}$

↳ Relative minus sign bwn. diagrams:

- If two diagrams are identical except for the exchange of two identical fermion lines, they have a relative minus sign.
- Add a minus sign if the fermion lines after interaction vertices of one diagram are odd permutations of another.

→ Computing squared matrix elements

↳ $e^- \psi \rightarrow e^- \psi$

$$\circ T = y^2 \bar{u}_s'(\vec{p}') \left[\frac{-\not{p} - \not{k} + m}{-s + m^2} + \frac{-\not{p} + \not{k}' + m}{-u + m^2} \right] u_s(\vec{p}) = \bar{u}_s(\vec{p}') A u_s(\vec{p})$$

$$\circ \text{For } \sigma, \Gamma, \text{ we want } |T|^2 = T T^* \rightarrow T^* = T^\dagger = (\bar{u}_s'(\vec{p}') A u_s(\vec{p}))^\dagger = u_s^\dagger(\vec{p}) A^\dagger \gamma_0 u_s'(\vec{p}')$$

$$\begin{aligned} \circ |T|^2 &= T^* T = (\bar{u}' A u)(\bar{u} A u') \\ &= \sum_{\alpha\beta\gamma\delta} \bar{u}'_\alpha A_{\alpha\beta} u_\beta \bar{u}_\gamma A_{\gamma\delta} u'_\delta \\ &= \sum_{\alpha\beta\gamma\delta} u'_\delta \bar{u}'_\alpha A_{\alpha\beta} u_\beta \bar{u}_\gamma A_{\gamma\delta} = \text{Tr}[(\bar{u}' u') A (\bar{u} u) A] \\ &= \bar{u}_s(\vec{p}) A u_s'(\vec{p}') \quad (\text{for this case}) \end{aligned}$$

↳ Casimir Trick:

- In practice, do not know incoming and outgoing spin states
- Average over initial spins, sum over final spins

in a collider exp. can

have 50% spin up, 50% spin down in beams

probability of spin up or spin down.

- spin averaged matrix element squared: $\langle |T|^2 \rangle = \frac{1}{2} \sum_{s,s'} |T|^2 = \frac{1}{2} \text{Tr} [(-\not{p}' + m) A (-\not{p} + m) A]$

$$\left(\sum_{s=\pm} u_s(\vec{p}) \bar{u}_s(\vec{p}) \right) = -\not{p} + m$$

↳ Trace theorems:

- want to evaluate trace of products of γ 's using $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$, $\gamma^2 = \mathbb{1}$, $\{\gamma^\mu, \gamma^5\} = 0$, $\text{Tr } \mathbb{1} = 4$

- $\text{Tr} [\text{odd \# of } \gamma^\mu\text{'s}] = 0$

$$\text{Tr} [\gamma^5 (\text{odd \# of } \gamma^\mu\text{'s})] = 0 \rightarrow \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

- $\text{Tr} [\gamma^\mu \gamma^\nu] = -4g^{\mu\nu} \Rightarrow \text{Tr} [\not{a} \not{b}] = -4(a \cdot b)$

$$\text{Tr} [\not{a} \not{b} \not{c} \not{d}] = 4[(a \cdot d)(b \cdot c) - (a \cdot c)(b \cdot d) + (a \cdot b)(c \cdot d)]$$

$$\text{Tr} [\gamma^5] = 0, \text{Tr} [\gamma^5 \gamma^\mu \gamma^\nu] = 0, \text{Tr} [\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = -4i\epsilon^{\mu\nu\rho\sigma}$$

→ Mandelstam variables:

$$s = -(p_1 + p_2)^2 = -(p_1' + p_2')^2 \Rightarrow p_1 \cdot p_2 = p_1' \cdot p_2' = -\frac{1}{2}(s - 2m^2)$$

$$t = -(p_1 - p_1')^2 = -(p_2 - p_2')^2 \Rightarrow p_1 \cdot p_1' = p_2 \cdot p_2' = \frac{1}{2}(t - 2m^2)$$

$$u = -(p_1 - p_2')^2 = -(p_2 - p_1')^2 \Rightarrow p_1 \cdot p_2' = p_2 \cdot p_1' = \frac{1}{2}(u - 2m^2)$$

- we can replace dot products in our $\langle |T|^2 \rangle$ with Mandelstam variables, giving simpler forms.

↳ cross sections: $\frac{d\sigma}{d\Omega_{cm}} = \frac{1}{64\pi^2 s} \frac{|\vec{p}_f|}{|\vec{p}_i|} \langle |T|^2 \rangle$